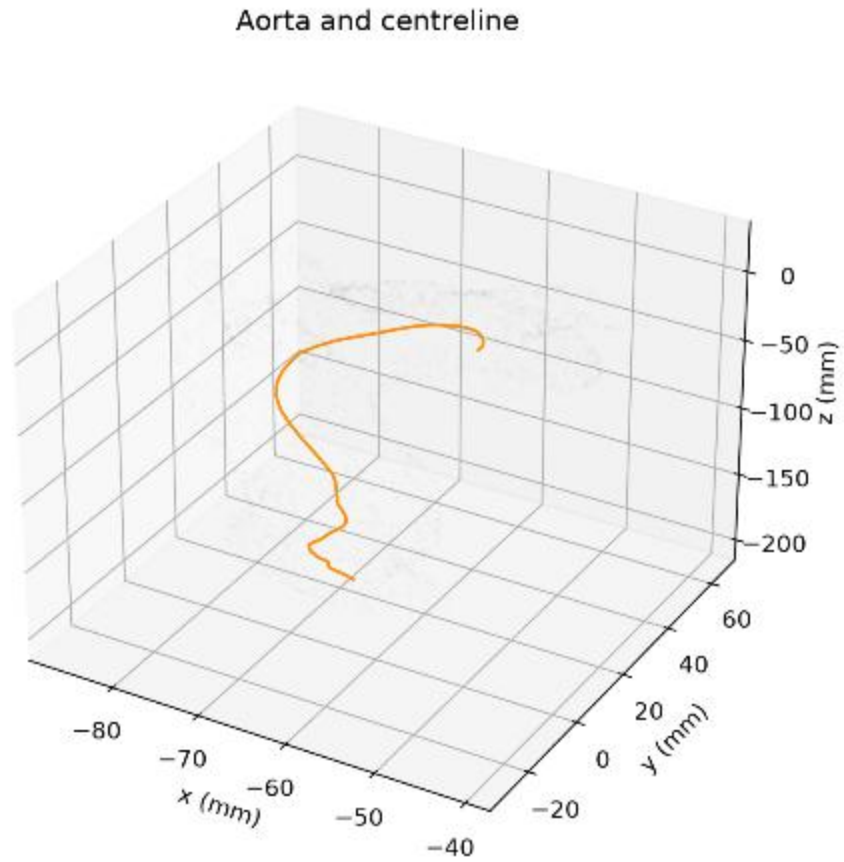
From Physics to Physiology
Project 2

1D flow assumptions in a healthy aorta

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The question and the 1D assumptions

How well do 1D assumptions hold in this aorta?

Each section becomes area A, flow Q and mean pressure $\bar{P}$.

- Mostly axial flow
- Uniform pressure within each section
- Fixed velocity-profile correction α
- Friction follows the assumed profile
- Mass balance along the vessel
- The full momentum balance

Measurements from the 3D simulation

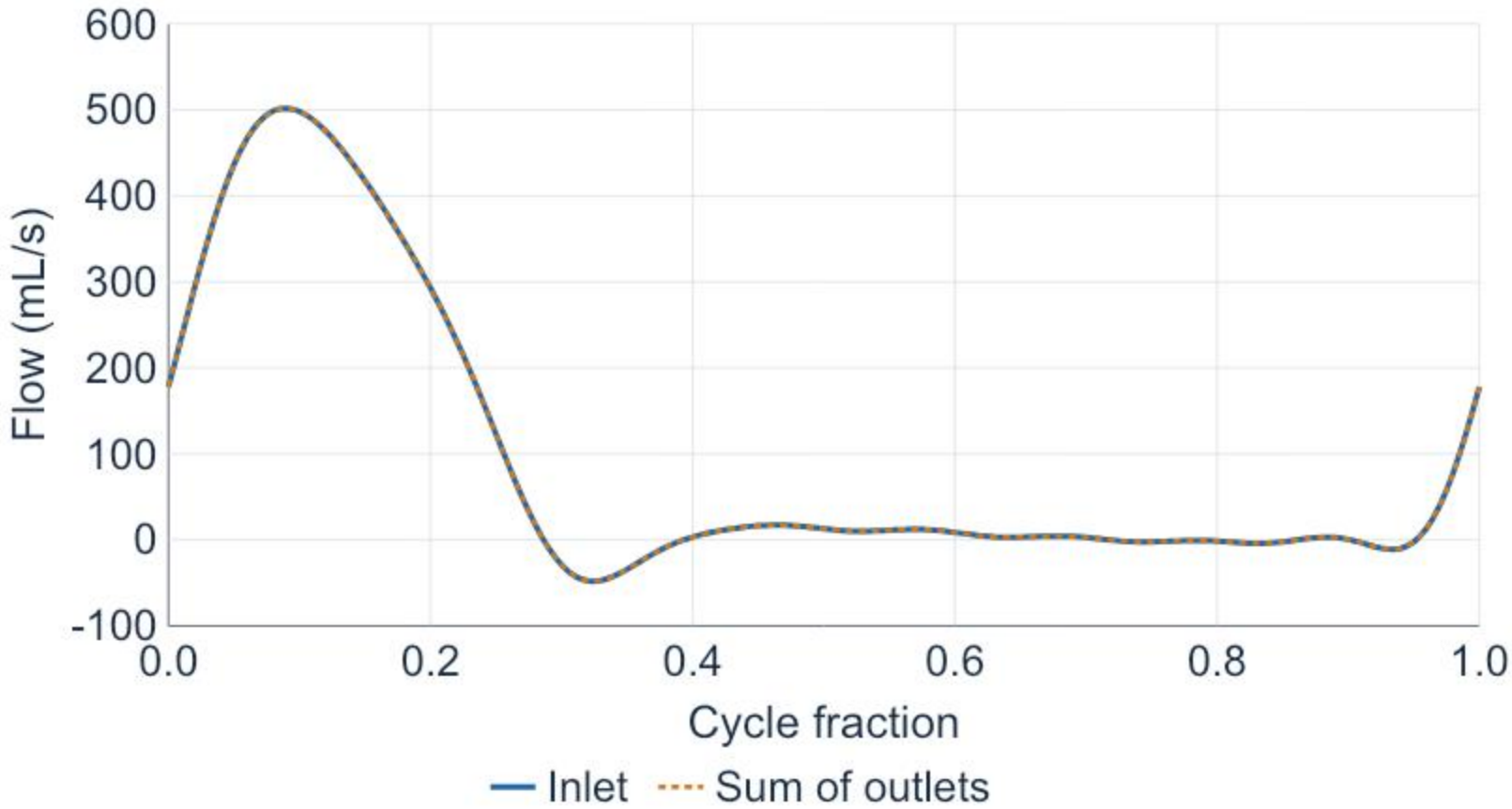
Existing rigid-wall data from the Vascular Model Repository

- Sections perpendicular to the centreline
- Integrate actual pressure and velocity with triangle-area weights
- Exclude branch regions from single-vessel derivatives

201 snapshots
47 usable sections
T = 0.937 s

$Q = \int u_s \, dA \qquad U = Q/A \qquad \bar{P} = (1/A) \int P \, dA$

The flow measurements conserve mass



Global mismatch

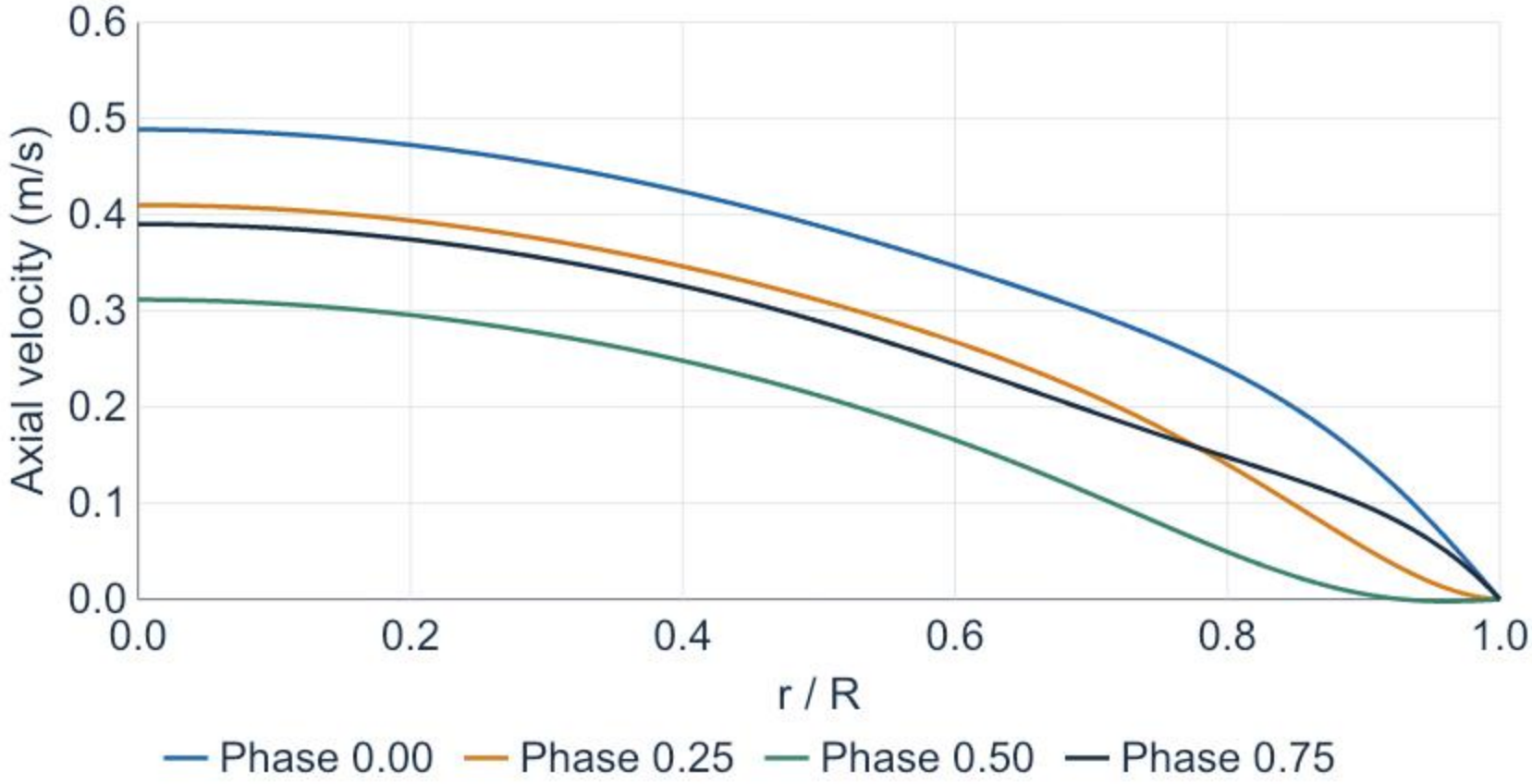
0.000022%

Local mismatch

0.35%

Maximum differences, scaled by peak inlet flow

A Womersley case checks the calculations



Known straight-tube solution

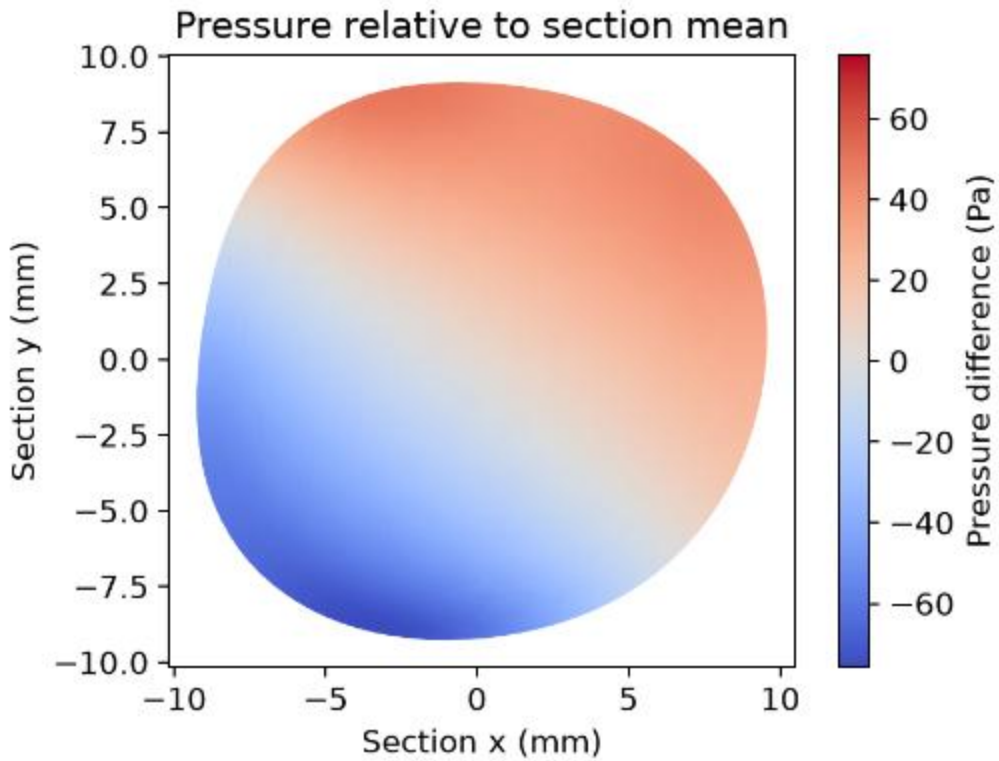
Mean-speed error
4.0×10^{-7} m/s

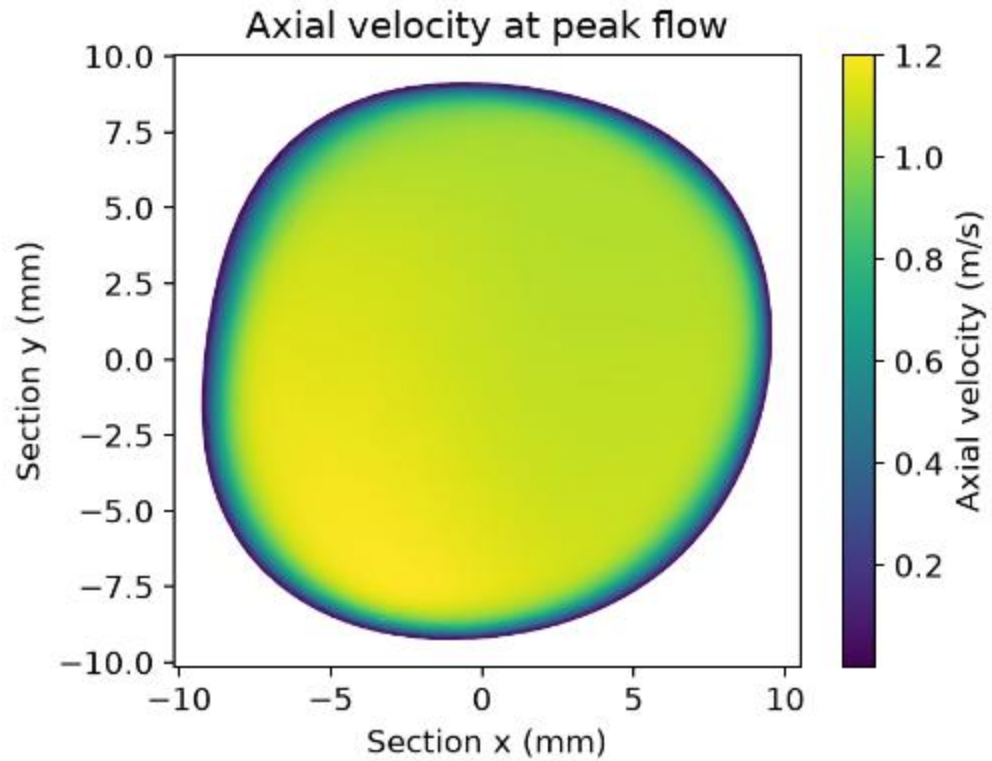
No-slip at the wall
Analytical balance holds

Reference calculation checks before interpreting the aorta

Pressure varies across the section

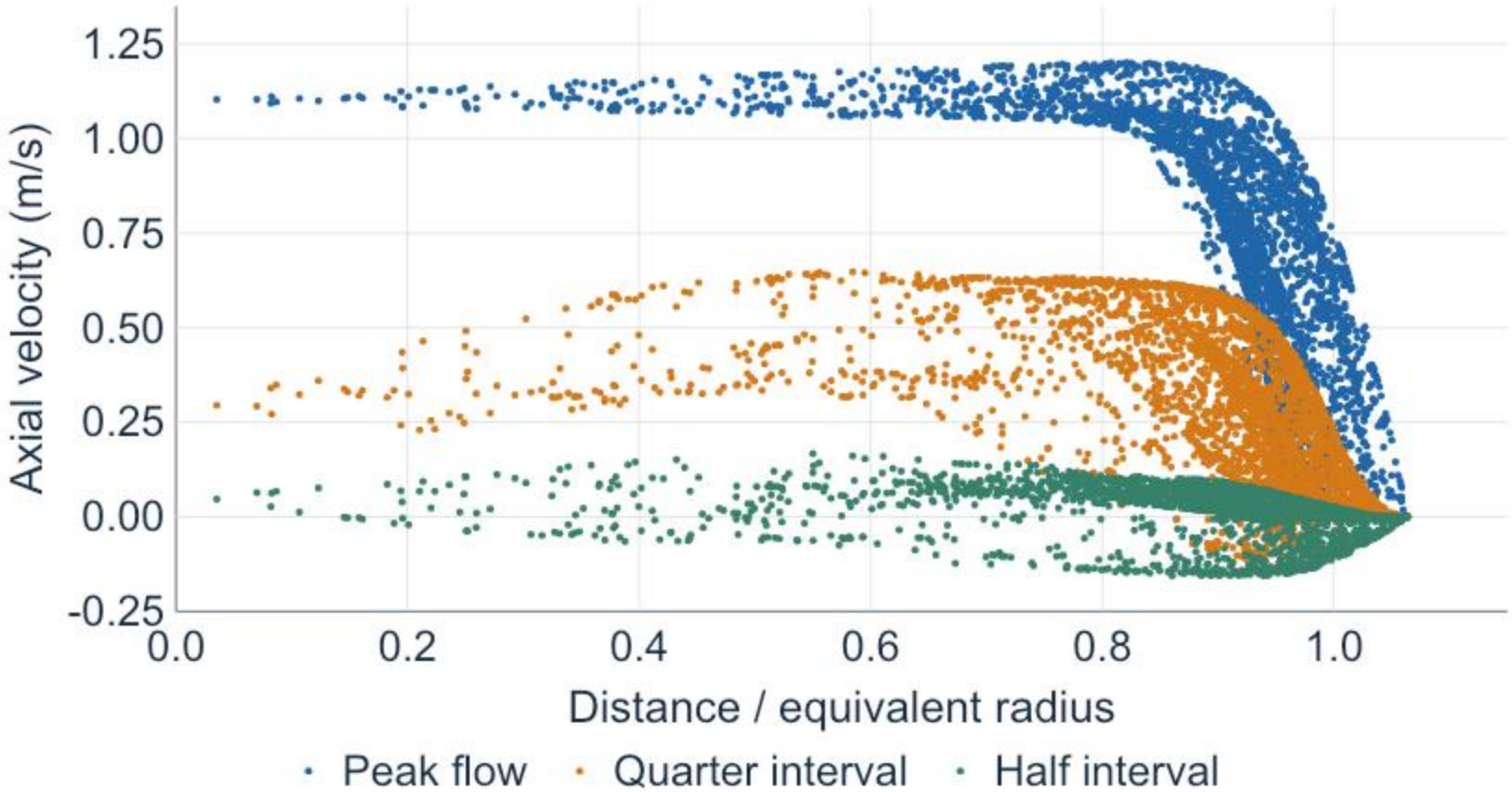
Section at 199 mm, at peak section flow





Pressure range: 126 Pa

The velocity profile changes with phase



At peak flow

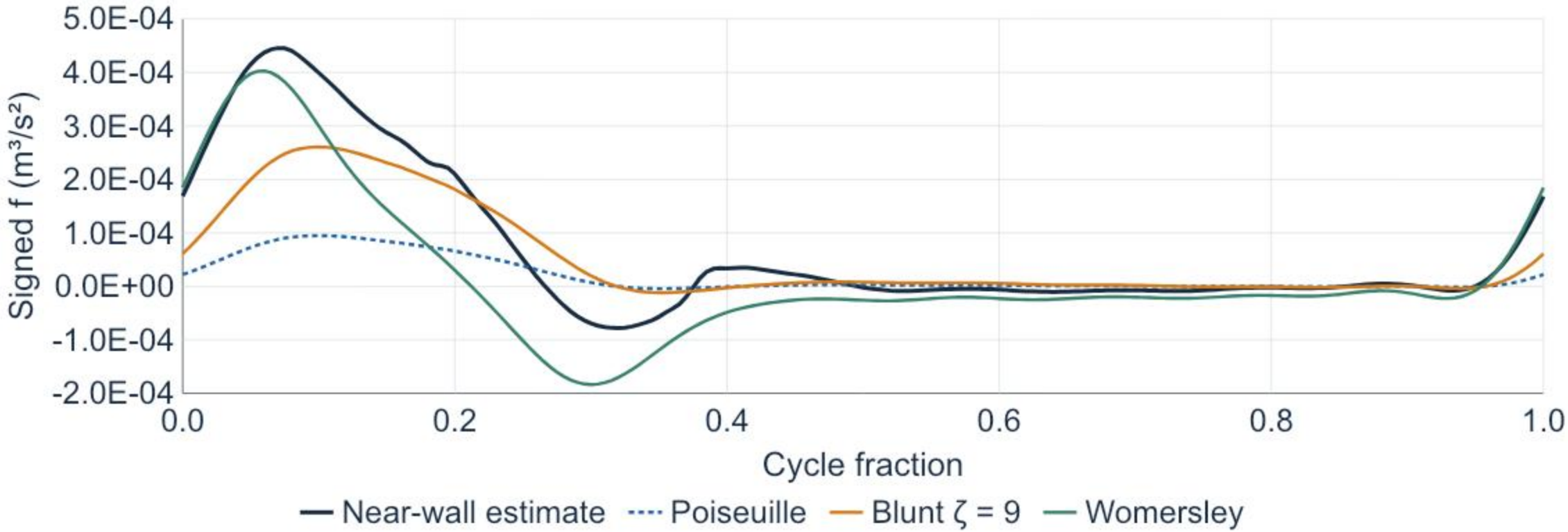
α = 1.06

In-plane RMS
7.1% of axial RMS

Point spread shows variation around the section

Reference profiles predict different friction

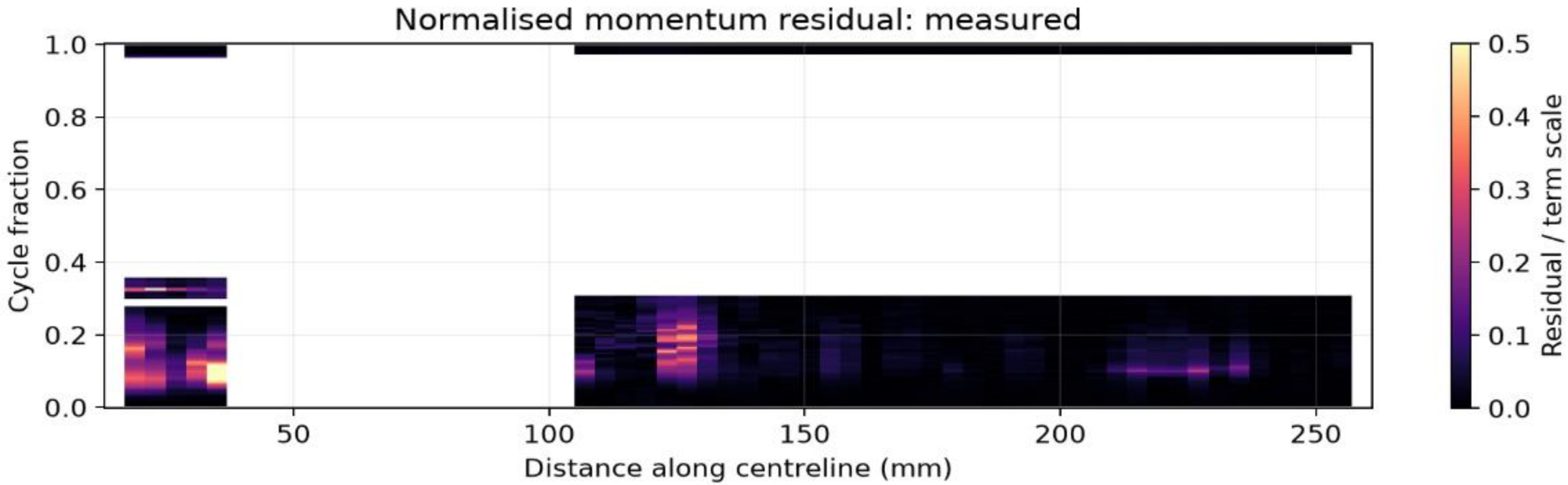
$f = K_r U$ Poiseuille: $K_r = 8\pi\nu$ Blunt $\zeta = 9$: $K_r = 22\pi\nu$



References use the same measured mean-flow history

The projected momentum balance

$$R = \partial Q/\partial t + \partial(\alpha Q^2/A)/\partial s + (A/\rho)\partial \bar{P}/\partial s + f$$



White regions: excluded or unavailable values. Summary uses common samples.

The closure comparison

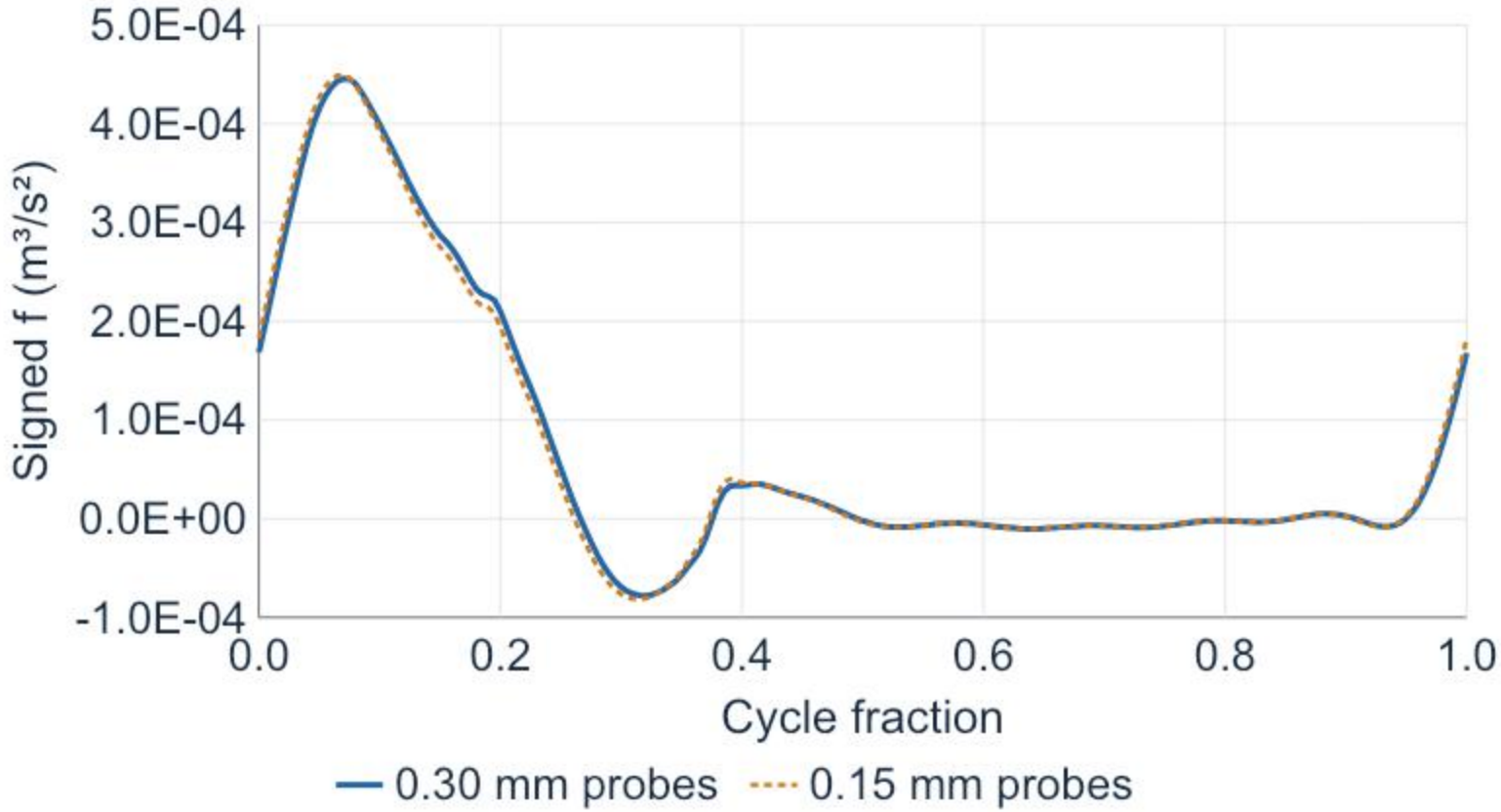
Normalised residual as a percentage of the term scale

Closure	Median	95th percentile
Poiseuille	4.6%	49.6%
Blunt $\zeta = 9$	4.4%	45.4%
Measured α and f	**1.2%**	**17.5%**

Same 2,878 station–time pairs for every closure

Measured corrections reduce the median. Larger residuals remain.

Limits of the analysis



- Rigid walls
- One anatomy and one cycle
- Near-wall sampling affects force
- No mesh-convergence evidence

Probe-depth comparison is a sensitivity check

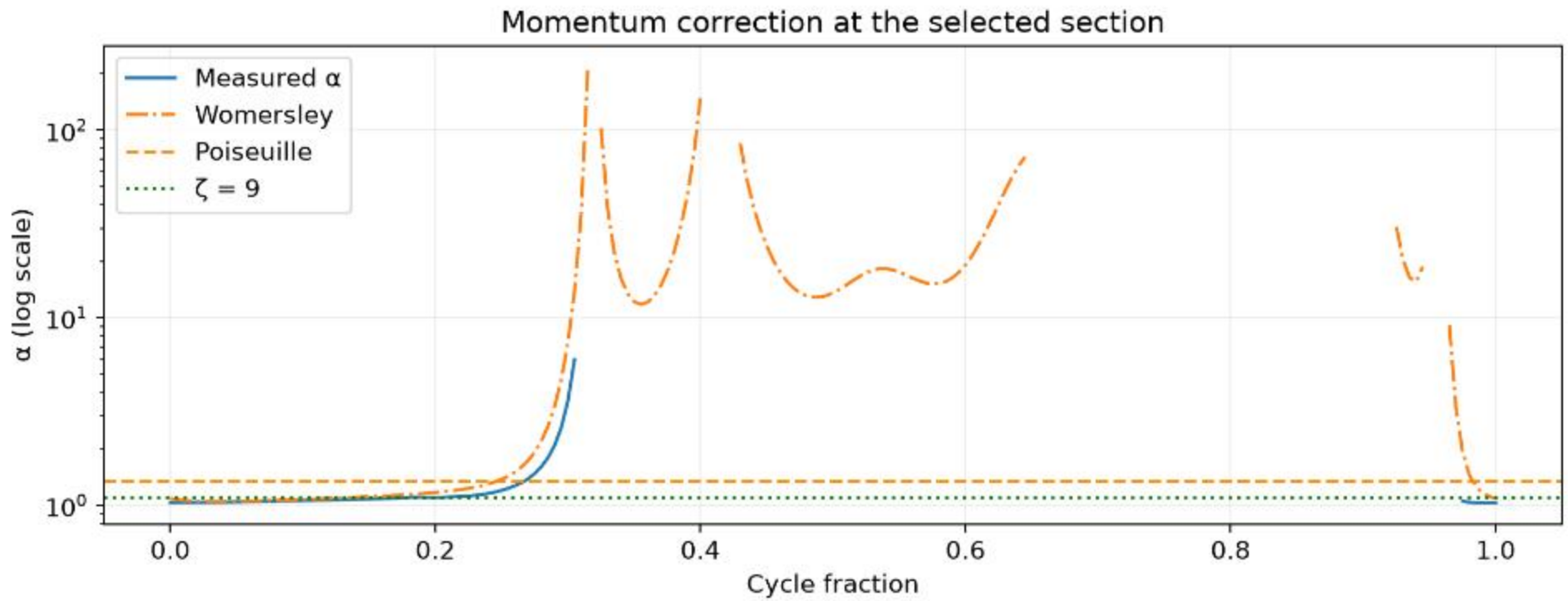
Conclusions

- Conservation supports consistent flow measurements.
- Measured profile and friction information reduces the typical residual.
- Position, phase and numerical limits still affect the result.

A 1D description can summarise bulk transport, with care in its closures.

Backup: profile correction over the cycle

Selected section. The vertical axis is logarithmic.



Actual α: 5% low-flow cutoff. Womersley reference: 1% cutoff.

Sources and reproducibility

Vascular Model Repository
Model 0012_H_AO_H and matching rigid-wall results
www.vascularmodel.com

Wilson, Ortiz and Johnson (2013)
Journal of Medical Devices 7(4), 040923
doi:10.1115/1.4025983

Project 2 brief and From Physics to Physiology teaching material

Notebook and included scripts: python main.py

AI assistance in drafting and code organisation